
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

on

AI-Mock Interview with Streaming Avatar

1. Introduction

1.1 Overview:

Picture a place where getting ready for interviews isn't about repeating lines. It's closer to talking with someone who listens. Built on smart language tech, this tool acts like a seasoned pro asking questions. As you answer, the flow changes - no rigid format holding it down. The pace breathes. Each question adjusts, shaped by what you say. Silence sits comfortably, not awkwardly forced. Out of nowhere, a new question pops up - much like how real interviews go. Talking with you live, an animated AI face dives into job-focused topics. Before anything else happens, your resume gets uploaded so the software can scan it, shaping queries that match what you've actually done. As you speak, voice tech turns words into text on the fly, letting the system weigh how clear and fitting each reply sounds. Hidden from view, several AI parts act separately - one plays the hiring manager, another checks technical depth, while one steps back to judge the whole picture. Later comes a summary pointing out what worked well, where things stumbled, along with spots worth refining. Think of it as a straightforward aid meant for those ready to rehearse interviews with purpose.

1.2 Purpose:

This work moves toward one clear aim — building an AI-powered system for practice job interviews. A place where training happens without heavy pressure. Safe. Quiet. Open to learning. Many candidates struggle during interview preparation. Not always because they don't know the subject. Often it's the lack of guidance, or feedback that never really tells the truth. This project tries to close that gap.

Inside the system, artificial intelligence drives the conversation. It listens. Then responds. Questions appear one after another, almost like a real interviewer thinking on the spot. Sometimes the tone feels strict, sometimes a bit relaxed. Responses are captured through

speech tools, turning spoken words into text. The system studies how answers are formed — not just what is said, but how it is said. Clarity matters. Confidence too. Small mistakes happen along the way. That's expected. Over time, the user starts noticing improvement. Answers become steadier. Thinking becomes quicker. Piece by piece, practice builds confidence.

Key purposes include:

- Simulate realistic interview scenarios using AI and conversational interaction.
- Generate personalized interview questions based on the user's resume, profile, or target role.
- Observe candidate responses during the session and analyze how clearly ideas are communicated.
- Break responses into smaller parts so the system can understand strengths and weak points better.
- Help users improve confidence and communication skills through repeated practice sessions.
- Demonstrate the practical use of Large Language Models and agent-based AI systems working together.

Errors appear here and there. That's fine. Learning grows slowly, step by step, like quiet progress you almost don't notice at first. But it happens.

2. Literature Survey

2.1 Existing Problem:

Years pass. New tools show up meant to get folks ready for interview chats. A few keep it basic. While some build tighter frameworks around practice. Yet nearly all chase one aim - getting applicants into a confident headspace ahead of live Q&A sessions. Looks solid when written down. Reality? Not always so clean.

Out there, lots of folks look up interview help on websites. Those sites spill out question after question, toss in sample replies, even advice on sitting straight or sounding sure. Sort of useful, at least for starters. People skim pages, jot notes, occasionally repeat phrases under their breath. Yet actual interviews? They twist without warning. Nothing stays fixed for long.

Some people learn by viewing clips of interview prep. Career advisors often post recordings online instead. Those show ways to build replies, start conversations, even manage tough moments. What you hear can help somewhat. Still, seeing it done feels nothing like living it firsthand. Practice talks with coaches happen too. Better results come when actual conversation kicks in. Someone poses questions, answers get shared, then comments arrive afterward. Trouble shows up anyway. Spending hours on these

meetings isn't always possible. Some learners lack people nearby willing to run practice rounds now and then. Money might also stand in the way. Regular feedback from experienced guides doesn't come easily for everyone.

Screen-based job interview helpers now pop up on certain sites. These show prompts while capturing replies via camera or mic. Progress? Perhaps. Yet most stick to rigid sets of questions. The back-and-forth lacks rhythm, leaving exchanges stiff. Flow breaks down fast.

Most current fixes tend to involve:

Online interview preparation websites with lists of common questions and suggested answers.

Career guidance videos explaining interview strategies and communication tips.

Fake job talks run by instructors, guides, or work advisors. Sessions shaped through practice rounds led by educators, supporters, or path planners. Role-play chats managed by coaches, helpers, or future planners. Trial conversations guided by tutors, backers, or direction specialists.

A fresh look at your resume, paired with guidance to shape answers. Coaching walks beside applicants, refining how they present experience. Support focuses on readiness, building confidence through practice. Help arrives when needed most - before interviews begin.

Some simple tools let people answer questions on video or by voice. These systems capture replies without live helpers watching. A camera or microphone picks up what users say during sessions. Recordings happen straight to software instead of face-to-face chats. Tech handles timing while applicants respond at their own pace.

Practice groups where students simulate interviews with each other.

Some fixes work okay. Yet a gap remains. Actual interviews go off script. Talk moves in ways you can't predict. Tension creeps in over time. Many current options fail at mimicking that messy, uneasy flow - the kind of moment when growth really kicks in.

2.2 Proposed Solution:

- The approach moves away from the usual preparation style. Instead of only reading questions or watching interview tips, it focuses on actual practice. Real interaction. Real responses. A practice space where mistakes can happen without pressure.
- The process begins when a user uploads their resume into the system. Artificial intelligence quickly studies the information. Skills, projects, work history — everything gets scanned. From there, the interview slowly starts to take shape.

- Advanced language models guide the conversation. Questions are not fixed beforehand. They adapt depending on what the candidate says. Sometimes the flow feels formal, other times a bit casual. Almost like a real interviewer thinking on the spot.
- An AI-generated avatar appears on screen and begins the interview. It asks questions, pauses, and reacts to responses. If an answer feels incomplete, another question may follow. The path of the interview keeps shifting based on the conversation.
- The system also listens carefully during the session. Speech recognition technology captures spoken responses and converts them into text. These responses are then analyzed for clarity, structure, and how well ideas are explained.
- Communication style matters here. The AI doesn't only check correct answers. It also looks at confidence, organization of thoughts, and how smoothly explanations are delivered.
- Behind the scenes, multiple AI agents quietly handle different tasks. One acts like an HR interviewer, focusing on background and experience. Another agent handles technical questions related to skills. A third evaluates the entire interaction and prepares feedback.
- After the interview session ends, the system produces feedback. First it highlights strengths. Then it points out weaker areas. Suggestions for improvement are also included so the candidate knows where to focus next.
- Repeating practice sessions helps candidates slowly build confidence. Each round improves clarity and response quality. Small progress appears over time.
- Mistakes during practice are expected. In fact, they help the learning process. With repeated sessions, thinking becomes faster and interview anxiety begins to fade.
- The system keeps track of previous practice sessions. Over time it begins to notice patterns in responses. Some answers improve, others still need work. This history helps the user see progress clearly. Slow improvement, but visible.
- Another useful part is the flexibility of the platform. Users can practice interviews for different job roles or industries. Technical roles, HR rounds, or general interviews. The questions shift depending on the chosen role, making each session slightly different.
- The platform is also available anytime, which makes practice easier. No need to wait for mentors or schedule mock interviews. A user can simply start a session whenever they feel ready. Late night practice. Quick revision before an interview. It all works.

3. Theoretical Analysis :

3.1 Block Diagram

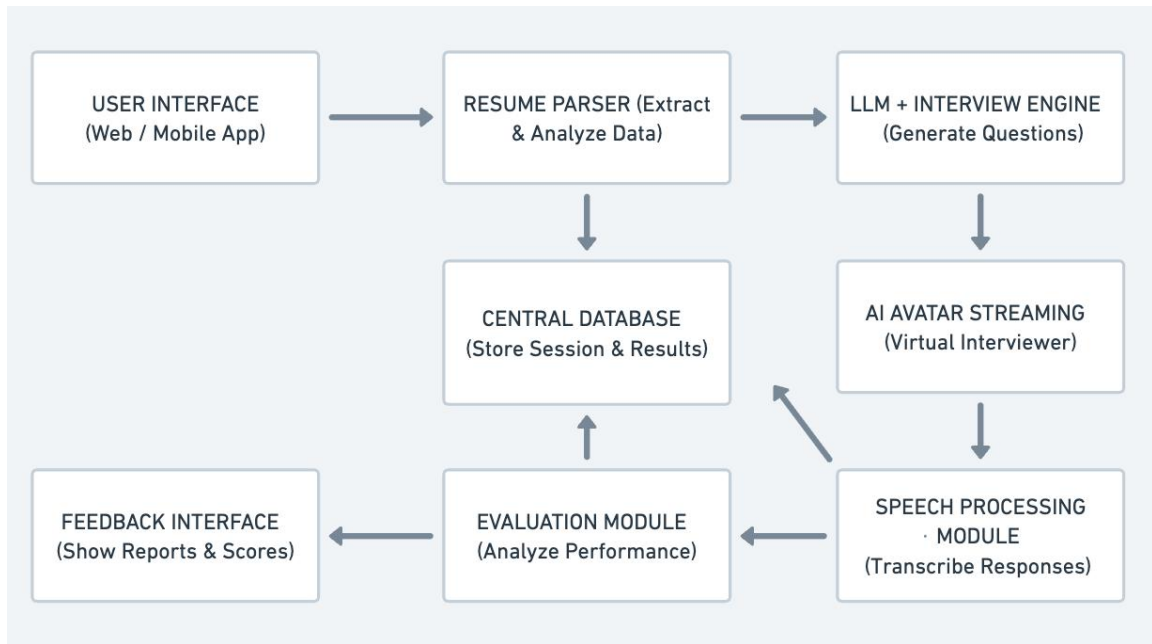


Fig. 1. Architecture of AI powered Mock Interview with Streaming Avatar

This flow ensures that users get accurate, real-time answers with minimal delay, while the backend handles all the complexity invisible.

3.2 Hardware Designing

Since this is a software-focused project, hardware needs are minimal:

- A laptop or desktop with internet access
- Minimum 8 GB RAM and i5 processor (recommended for smooth development)
- Optional: Server for hosting chatbot (can use cloud services like AWS or Firebase)

3.2.1 Software Designing

Component	Frameworks
Programming language	Python (backend and NLP integration)
Framework	Flask or Django (for web interface), TensorFlow or spaCy(for NLP)
Database	MySQL , Firebase (to store collage data)
Frontend Tools	HTML , CSS , JavaScript (for user interface)
APIs	OpenAI API

Table 1. Software Designing

4. Applications

This setup helps out wherever prepping for interviews counts. Students aren't the only ones who benefit. Workers looking for new roles might lean on it too. So could schools that teach job skills. Even hiring teams inside firms may find a way to plug in. Imagine it as a space to test yourself. Where stumbling once or twice is part of moving forward. Practice without the weight of actual job searches. Step by step, comfort grows. Not rushed, just moving forward.

Now here's something odd - knowing a topic inside out doesn't always help when the camera turns on. The moment starts, voice shifts. Ideas freeze mid-air, refusing to land. This setup lets anyone step back into that instant, repeat it, sit with the discomfort. Slowly, the tightness eases. Familiarity creeps in. Strength builds where fear once sat.

Applications include:

- **Student Interview Practice:**

Facing job interviews on campus? Students log in to rehearse answers ahead of time. Depending on the career path picked, different prompts show up. Hard ones pop up now and then. Easy ones too. Each round builds confidence without feeling forced. Useful every single time.

- **Job Seeker Preparation**

Starting fresh somewhere usually means brushing up on interviews, especially when time has passed. One thing that helps is having room to try out responses, line up ideas, while also hearing what comes across in speech.

- **Career Training Institutes**

Practice sessions on the platform let coaching centers support learners without assigning a teacher each time. Training institutes find it useful when guiding pupils through mock interviews. Some schools rely on this setup to save staff hours. Learners repeat drills at their own pace while mentors step in only when needed. The system works well during busy enrollment periods. Staff redirect energy to complex tasks instead of routine coaching. Institutions maintain consistency across batches using automated tools. Pupils gain confidence by trying again after feedback loops. Not every exercise demands live supervision anymore. This method fits programs focused on job readiness skills.

- **Skill Assessment Tool**

Early on, some firms or courses could check how well someone communicates or handles challenges using this method. It helps them see practical thinking before moving forward.

- **Communication Skill Improvement**

A few people find understanding things easy. Speaking plainly trips them up instead. Practicing interview questions builds confidence. It sharpens how thoughts come out under stress.

- **Self-Learning Platform**

Any hour works for running through interview questions. Even if it is two in the morning. Or five past six. It does not make a difference. Waiting there, turned on, every single time someone shows up to go again.

Little changes show up after a while. Words come out more plainly. Sureness builds bit by bit. Then talking to interviewers feels less heavy.

REFERENCES

- I. Sagare, S. S., et al. **“Review Paper on AI-Driven Mock Interview System.”** Journal of Advances in Artificial Intelligence and Financial Research, 2025. This paper explains a smart system that simulates interviews using NLP and speech analysis to evaluate communication skills and provide automated feedback.
- II. Ayush Modanval, et al. **“Mock Interview Platform with a Real-Time AI Voice Agent.”** International Journal of Scientific Research in Engineering and Management, 2025. The system uses conversational AI, voice interaction, and adaptive questioning to simulate realistic interviews.
- III. Dayal, A. **“AI-Driven Mock Interviews: A Catalyst for Enhanced Interview Preparation.”** Journal of Engineering Education and Technology, 2025. The study analyzes how AI-based mock interview tools improve candidate confidence and readiness.
- IV. **“AI-Powered Mock Interview Preparation.”** International Journal for Modern Trends in Science and Technology, 2025. The research describes a platform architecture using React frontend, FastAPI backend, and AI models to generate interview questions and feedback.
- V. **“AI-PrepMate: AI-Assisted Mock Interview and Feedback System.”** International Journal of Advanced Research in Computer and Communication Engineering, 2025. The paper introduces a platform that records interview responses and provides automated performance

analysis.

VI. Nguyen, T. T. H., N. H. Nguyen and T. H. Thruong “SimInterview: Large Language Model-Based Simulated Interview Training System.” arXiv Research Paper, 2025. The study demonstrates how LLM-driven agents can conduct personalized interview simulations and provide feedback to improve interview readiness

GitHub Repository link: <https://github.com/Yuvi25Jain/AI-Mock-Interview-with-Streaming-Avatar>

Guided By:

Group Members:

Internal Mentor:

Siya Purohit (0827IT231134)

Tulja Vishwarkarma (0827IT231142)

Prof. Pawan Kumar Gupta

Yukti Baldua (0827IT231153)

External Mentor:

Yuvanshi Bhalawat (0827IT231154)

Ms Kalpana Dave